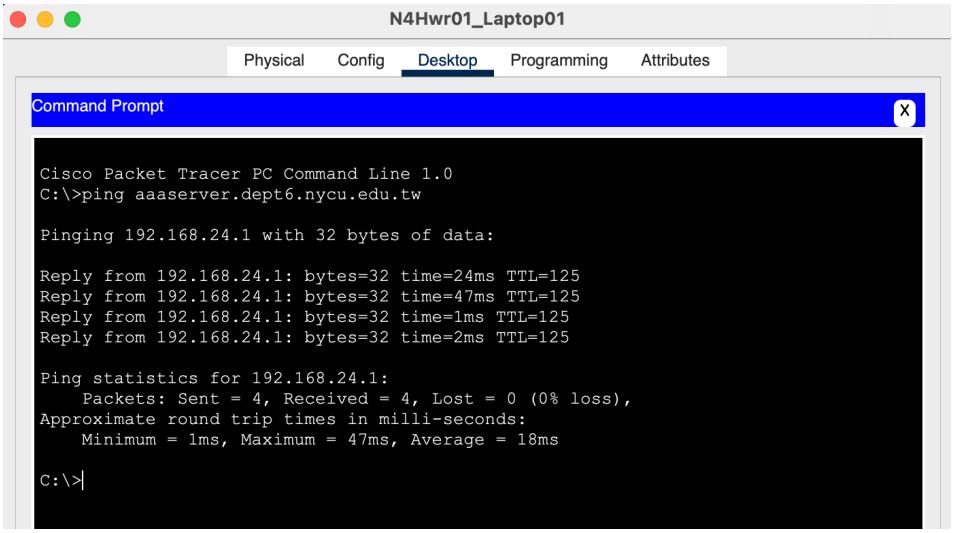

Fig.P1_t1

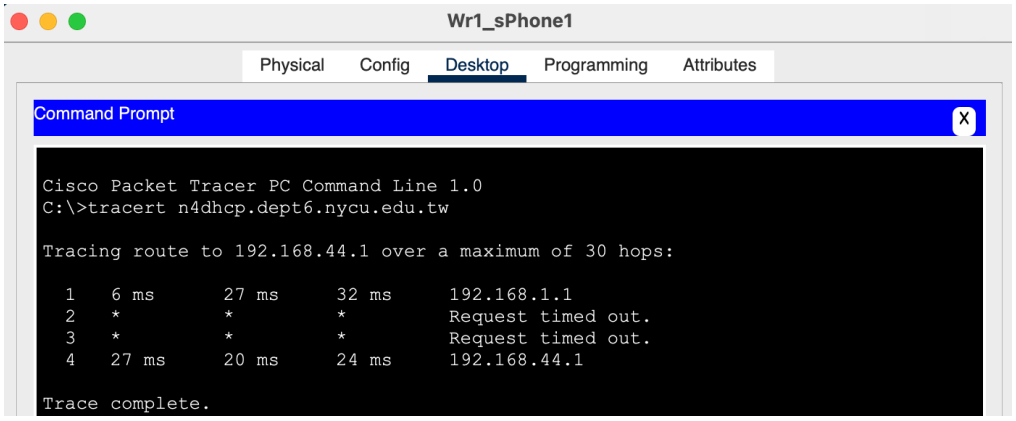

Fig.P1_t2a

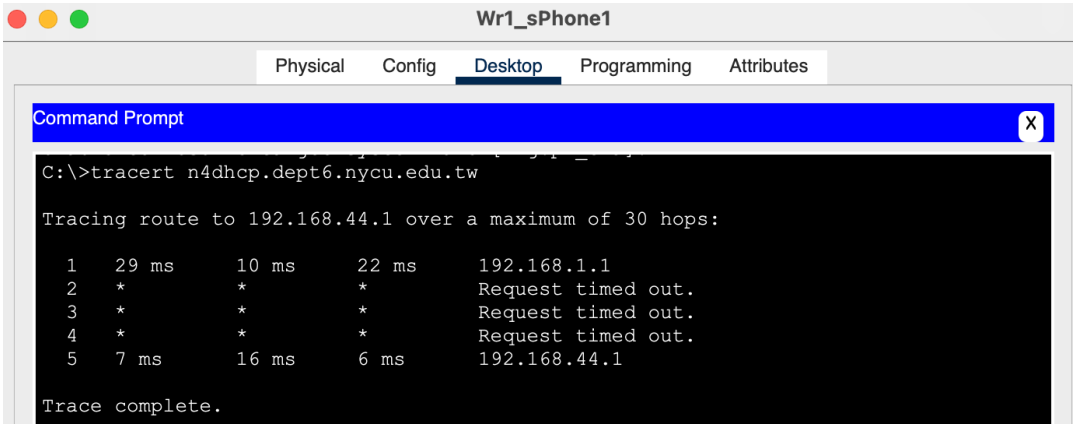

Fig.P1_t2b

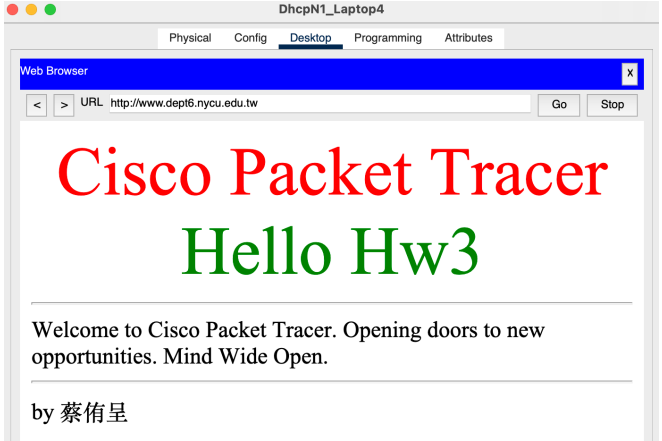

Fig.P1_t3

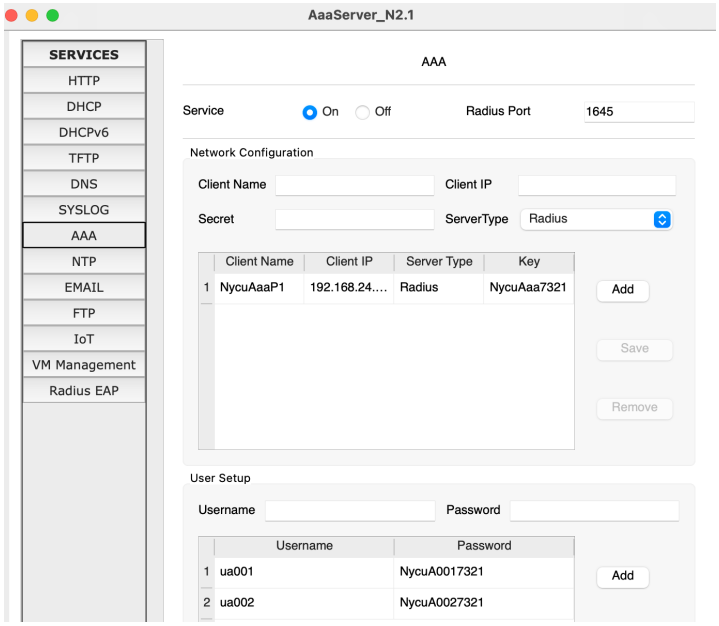

Fig.P1_s1

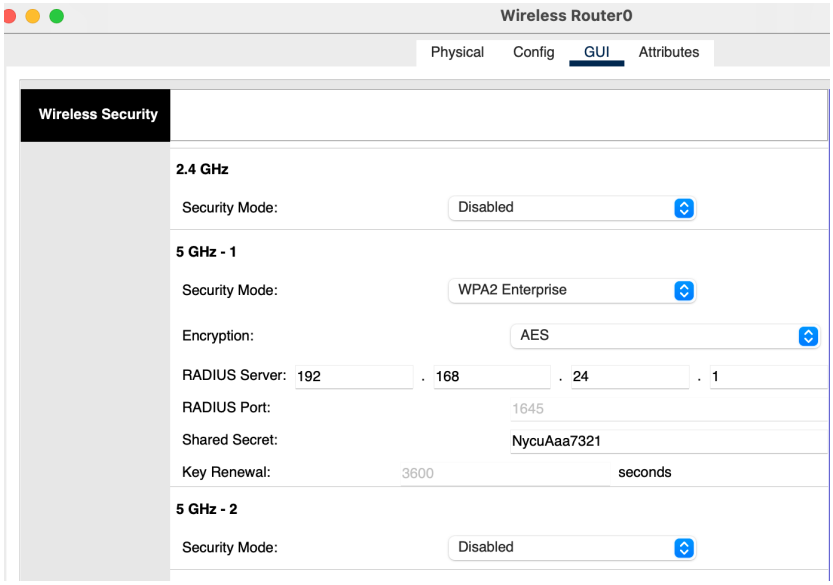

Fig.P1_s2

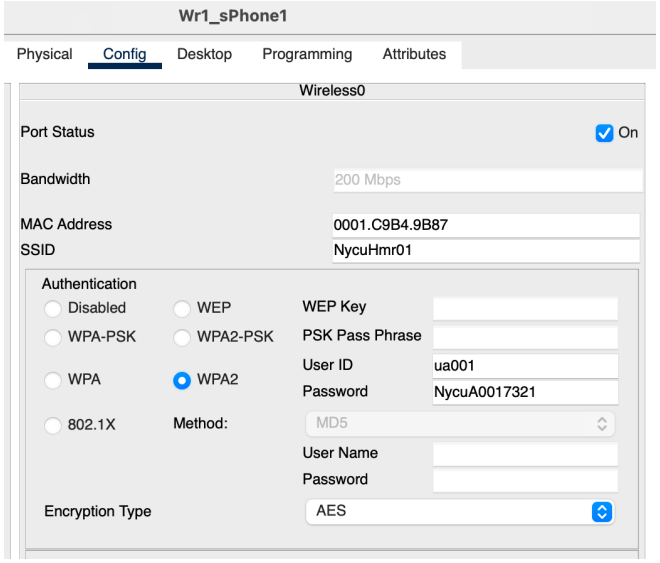

Fig.P1_s3

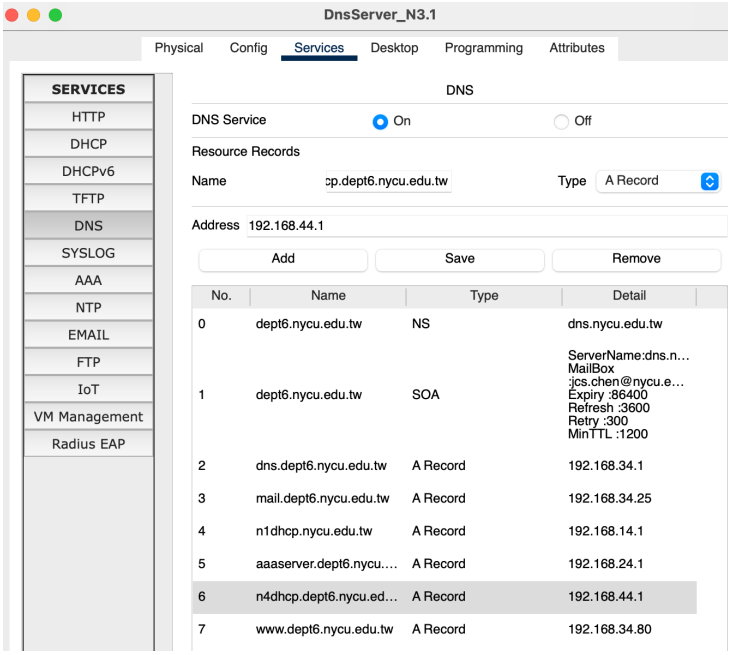

Fig.P1_s4

```
Device Name: Router_R1
Device Model: 2911
Hostname: Router

Port                    Link   VLAN   IP Address
GigabitEthernet0/0      Up     --     192.168.104.254/24
GigabitEthernet0/1      Up     --     <not set>
GigabitEthernet0/1.1    Up     --     192.168.14.254/24
GigabitEthernet0/1.2    Up     --     192.168.24.254/24
GigabitEthernet0/1.20   Up     --     <not set>
GigabitEthernet0/2      Down   --     <not set>
Serial0/3/0             Up     --     192.168.114.1/24
Serial0/3/1             Down   --     <not set>
```

Fig.P1_d1

```
Device Name: Wireless Router0
Device Model: HomeRouter-PT-AC

Port                Link   IP Address          MAC Address
Internet            Up     192.168.24.201/24   0004.9A44.9A01
Virtual-Access1     Up     <not set>           00D0.BC0A.CBB9
Dialer1             Up     <not set>           0040.0B76.8B69
LAN                 Up     192.168.1.1/24      0000.0C45.D006
Internet            Up     --                  0004.9A44.9A01
GigabitEthernet 1   Up     --                  0004.9A44.9A02
GigabitEthernet 2   Down   --                  0004.9A44.9A03
GigabitEthernet 3   Down   --                  0004.9A44.9A04
GigabitEthernet 4   Down   --                  0004.9A44.9A05
Wireless 1          Up     --                  0004.9A44.9A06
Wireless 2          Up     --                  0004.9A44.9A07
Wireless 3          Up     --                  0004.9A44.9A08
Wireless 4          Up     --                  0004.9A44.9A09
Wireless 5          Up     --                  0004.9A44.9A0A
Wireless 6          Up     --                  0004.9A44.9A0B
Wireless0/0         Up     --                  0004.9A44.9A0C
```

Fig.P1_d2

```
happyoyomg@Manager-node:~/Hw/Hw03$ sudo docker service create --name swarm_cluster --repl
icas=2 -p 8081:80 nginx-server:latest
image nginx-server:latest could not be accessed on a registry to record
its digest. Each node will access nginx-server:latest independently,
possibly leading to different nodes running different
versions of the image.

cax6gcscxvx8oale3z0pcldol
overall progress: 2 out of 2 tasks
1/2: running
2/2: running
verify: Service cax6gcscxvx8oale3z0pcldol converged
happyoyomg@Manager-node:~/Hw/Hw03$ sudo docker server ls
docker: unknown command: docker server

Run 'docker --help' for more information
happyoyomg@Manager-node:~/Hw/Hw03$ sudo docker service ls
ID             NAME            MODE         REPLICAS   IMAGE                 PORTS
cax6gcscxvx8   swarm_cluster   replicated   2/2        nginx-server:latest   *:8081->80/t
cp
happyoyomg@Manager-node:~/Hw/Hw03$ sudo docker service ps swarm_cluster
ID             NAME              IMAGE                 NODE           DESIRED STATE   CUR
RENT STATE               ERROR     PORTS
uevqkyzy65fl   swarm_cluster.1   nginx-server:latest   Worker-node1   Running         Run
ning about a minute ago
ymvvrej4p50u   swarm_cluster.2   nginx-server:latest   Manager-node   Running         Run
ning about a minute ago
```

Fig.P2_t1a

```
happyoyomg@Worker-node1:~$ for i in {1..6}; do curl http://192.168.56.2:8081/; e
cho; done
Nginx on Worker-node1 (113705056)

Nginx on Manager-node (113705056)

Nginx on Worker-node1 (113705056)

Nginx on Manager-node (113705056)

Nginx on Worker-node1 (113705056)

Nginx on Manager-node (113705056)
```

Fig.P2_t1b

```
happyoyomg@Manager-node:~/Hw/Hw03$ sudo docker service scale swarm_cluster=3
swarm_cluster scaled to 3
overall progress: 3 out of 3 tasks
1/3: running
2/3: running
3/3: running
verify: Service swarm_cluster converged
happyoyomg@Manager-node:~/Hw/Hw03$ sudo docker service ls
sudo docker service ps swarm_cluster
ID             NAME            MODE         REPLICAS   IMAGE                 PORTS
cax6gcscxvx8   swarm_cluster   replicated   3/3        nginx-server:latest   *:8081->80/t
cp
ID             NAME              IMAGE                 NODE           DESIRED STATE   CUR
RENT STATE               ERROR     PORTS
uevqkyzy65fl   swarm_cluster.1   nginx-server:latest   Worker-node1   Running         Run
ning 4 minutes ago
ymvvrej4p50u   swarm_cluster.2   nginx-server:latest   Manager-node   Running         Run
ning 4 minutes ago
tv6f82mk2opo   swarm_cluster.3   nginx-server:latest   Worker-node2   Running         Run
ning about a minute ago
```

Fig.P2_t2a

```
happyoyomg@Worker-node1:~$ for i in {1..9}; do curl http://192.168.56.2:8081/; e
cho; done
Nginx on Worker-node2 (113705056)

Nginx on Worker-node1 (113705056)

Nginx on Manager-node (113705056)

Nginx on Worker-node2 (113705056)

Nginx on Worker-node1 (113705056)

Nginx on Manager-node (113705056)

Nginx on Worker-node2 (113705056)

Nginx on Worker-node1 (113705056)

Nginx on Manager-node (113705056)
```

Fig.P2_t2b

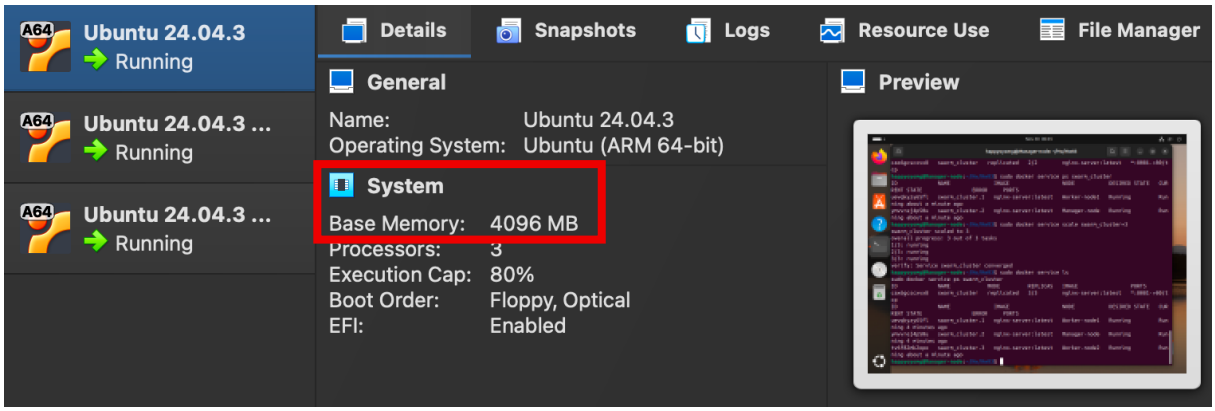

Fig.P2_Vbox